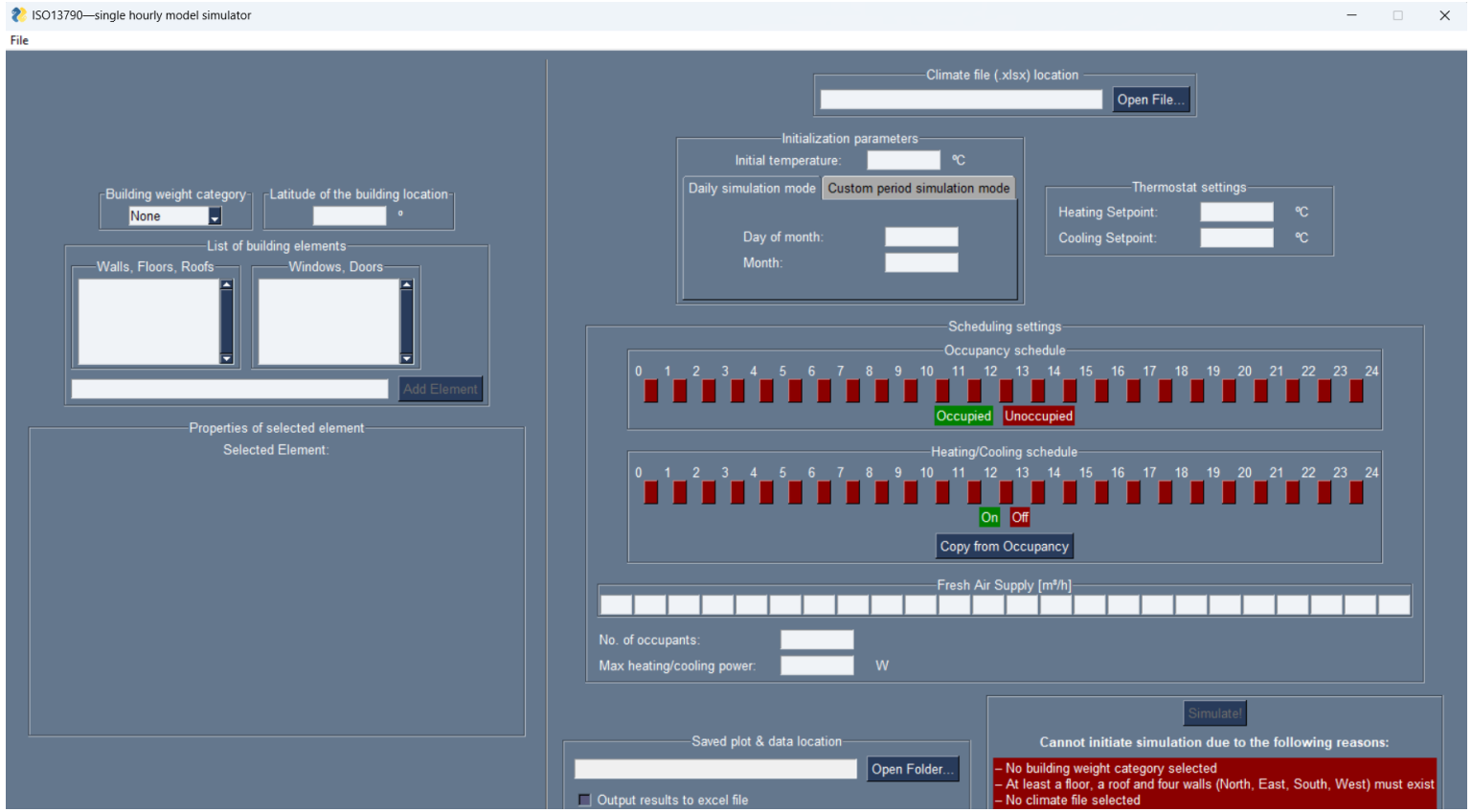
ISO13790-simple hourly method graphical environment, version 0.6.1

Documentation

Introduction



The software is suitable for building simulations with a 1-minute time step, from daily to annual level.

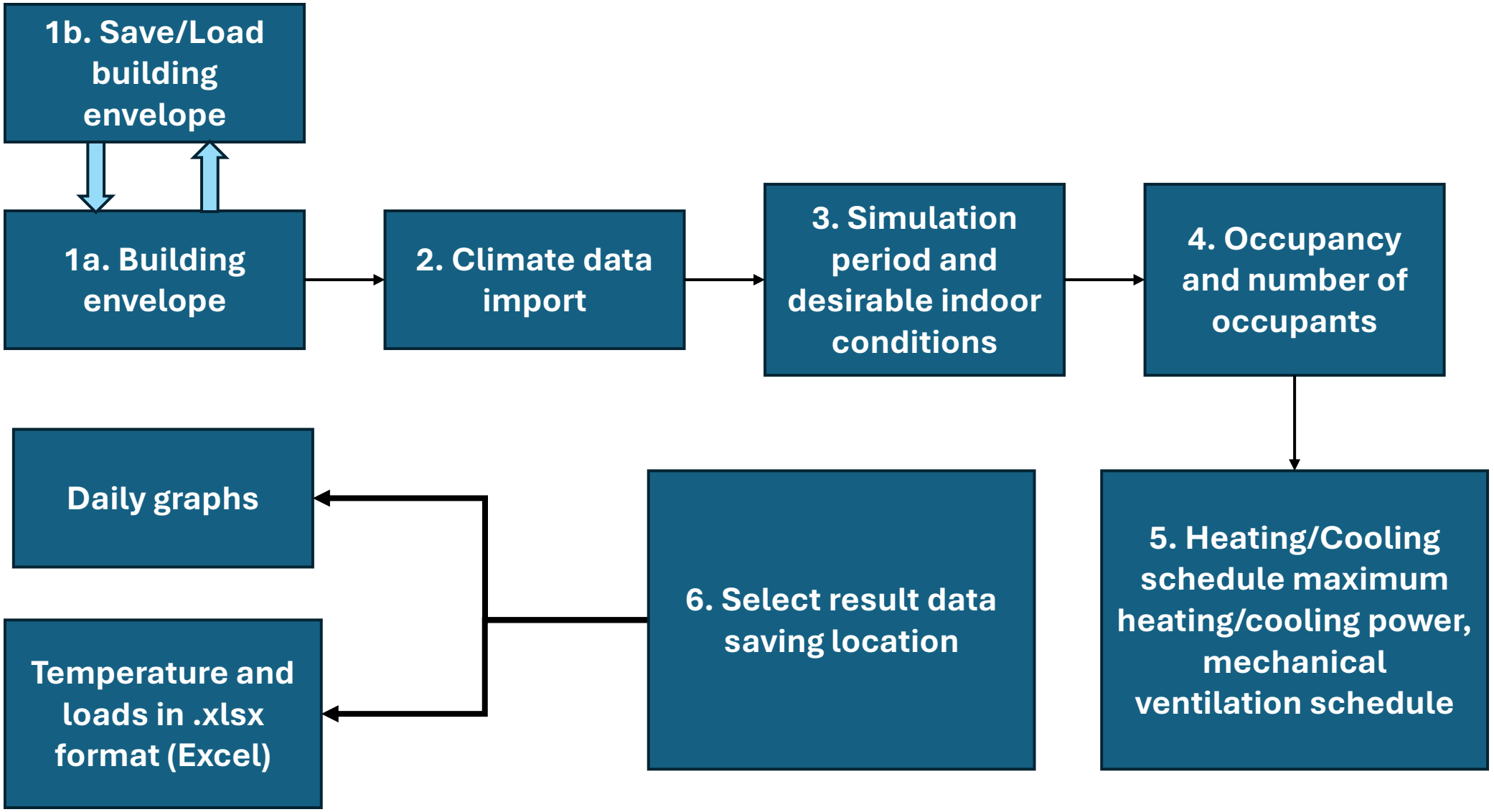
In the program it is possible to simulate:

- Spaces in residential and office buildings, which are considered as a single climatic zone surrounded by the external environment.
- Thermal loads:
  - Losses due to thermal transmittance of building elements
  - Losses due to ventilation
  - Solar gains
  - Gains due to the presence of occupants in the space, assuming residential or office activity

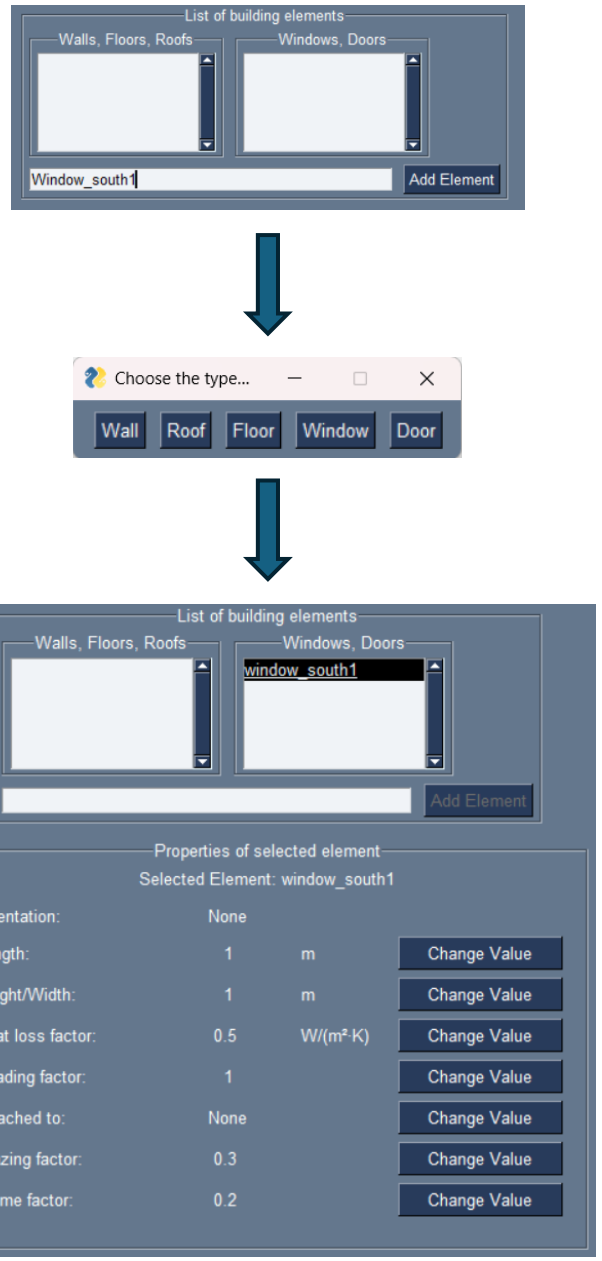
Currently, the program cannot simulate:

- Roofs
- Walls in contact with unheated spaces

Simulation steps



1a. Building envelope

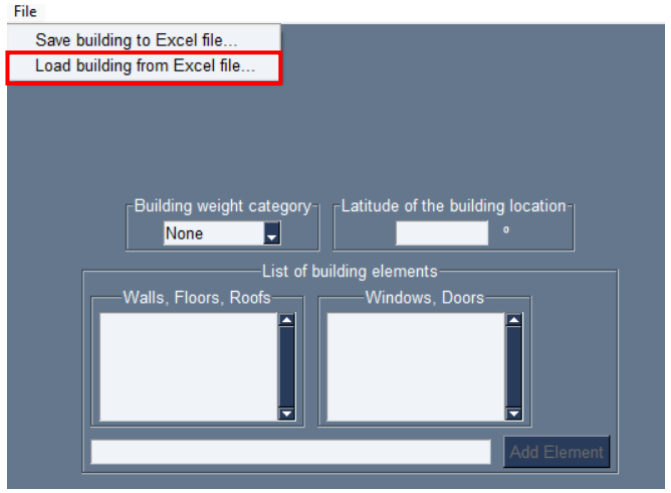


Ορίζεται κατηγορία βάρους κτιρίου (βάσει ISO EN13790) και γεωγραφικό πλάτος τοποθεσίας κτιρίου

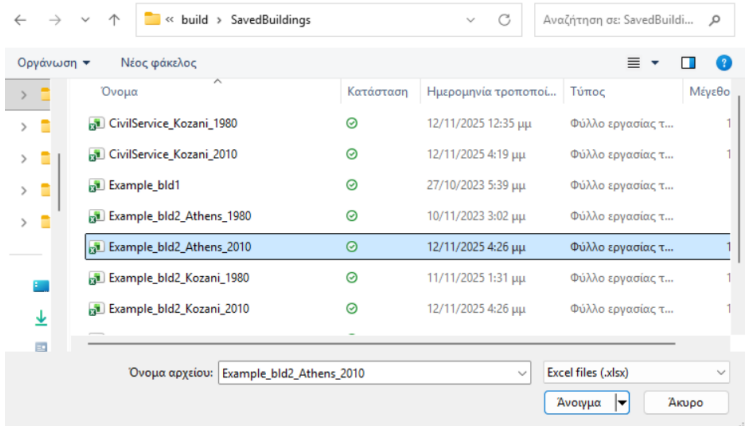
Ορίζονται τα δομικά στοιχεία του κτιρίου, επιλέγοντας το είδος τους

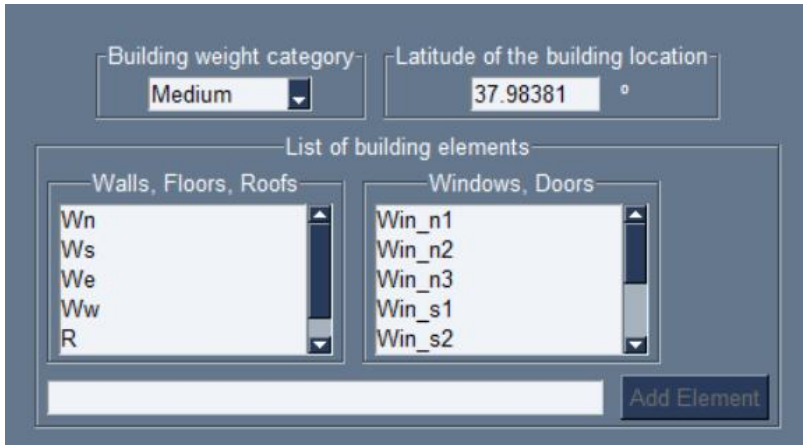
Αφού οριστεί το στοιχείο, καταχωρείται σε μια από τις δύο λίστες δομικών στοιχείων (σύμφωνα με το σχήμα αριστερά). Επιλέγοντας το στοιχείο είναι δυνατή η τροποποίηση των ιδιοτήτων του

1b. Save/Load Building Envelope



Load





Save

By selecting File → Save building to Excel file, it is possible to save the building envelope characteristics to an external Excel file as a building file for future use.

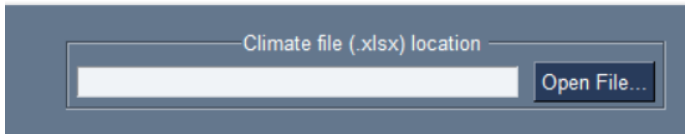
Load

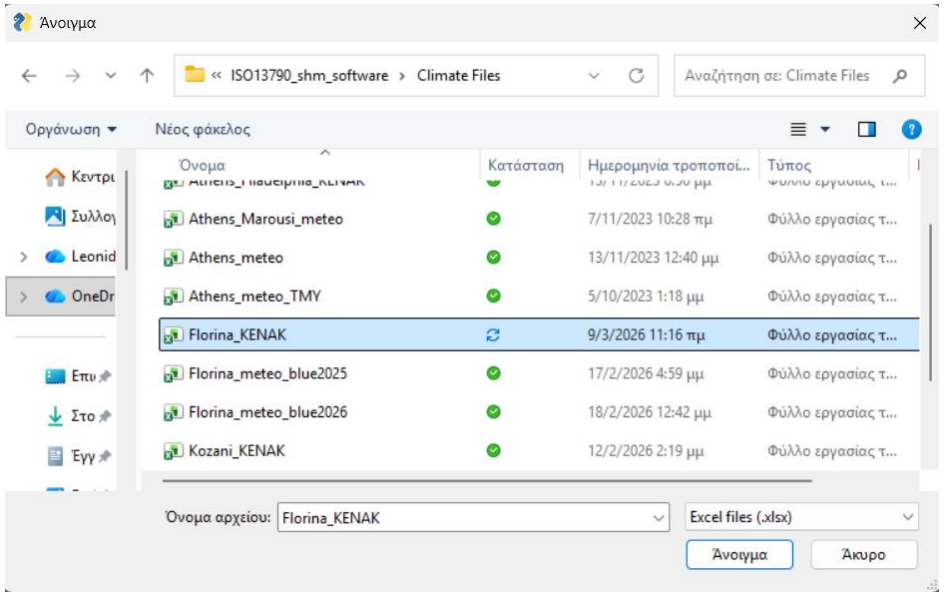
With File → Load building from Excel file, information from a saved building file is loaded into the software.

<u>Caution</u>

- For proper operation, the files generated through Save should not be modified directly, but only through the software.
- For the Load function, use files that were generated through Save, or copies of those files.

2. Climate data





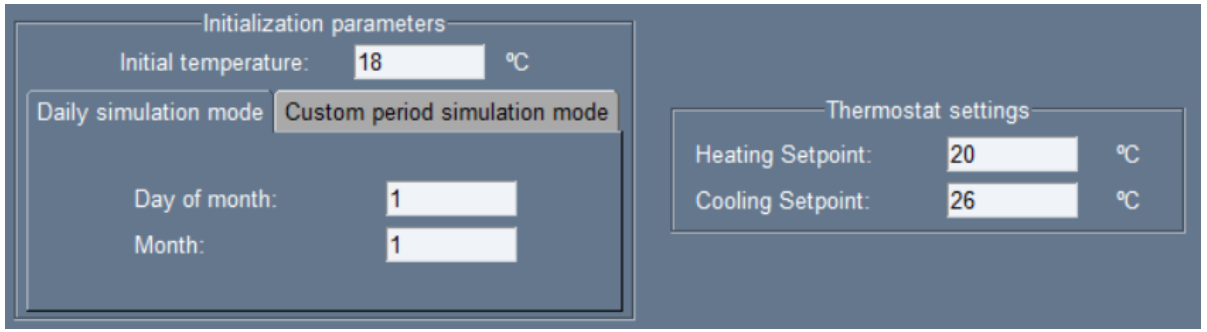
Although climate data files can be loaded from any location on the computer, the .xlsx files must follow a specific format.

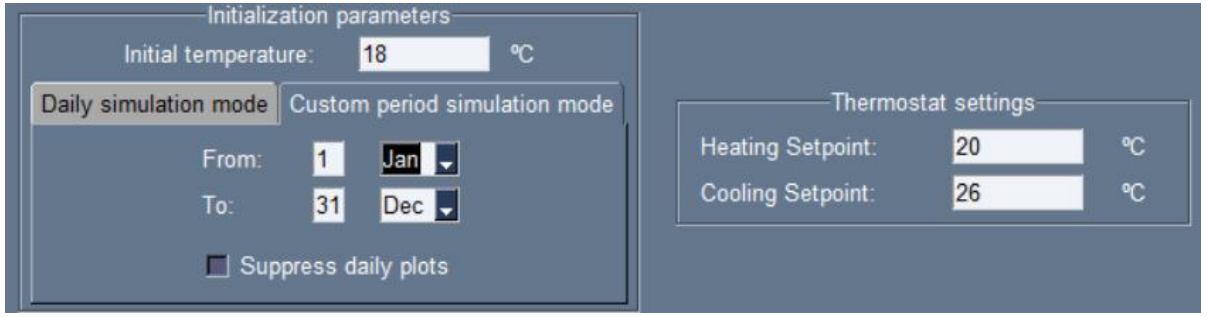
For the creation of new datasheets, use the file “Florina_KENAK.xlsx” in the climate data folder as a template.

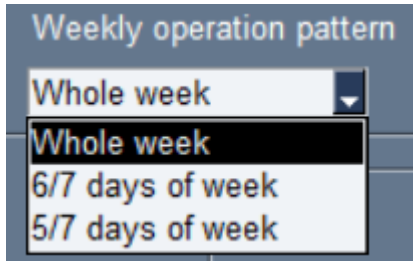
Template contains:

- Number of day of the year
- Month
- Day of Month
- Hour of Day (24h format)
- Ta – outdoor temperature [°C]
- G Global horiz [W]
- G Diff horiz [W]
- G Beam horiz [W]
- G 90 Deg, South [W]
- G 90 Deg, East [W]
- G 90 Deg, West [W]
- G 90 Deg, North [W]
- G diff 90 Deg, South [W]
- G diff 90 Deg, East [W]
- G diff 90 Deg, West [W]
- G diff 90 Deg, North [W]
- G beam 90 Deg, South [W]
- G beam 90 Deg, East [W]
- G beam 90 Deg, West [W]
- G beam 90 Deg, North [W]
- G direct normal [W]

3. Simulation period and desirable indoor conditions







Here the following are selected:

- The initial building temperature
- The simulation period

Daily simulation

- Daily temperature and energy charts are produced.

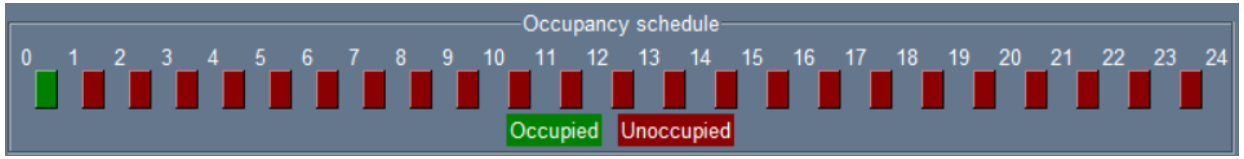
Custom simulation

- In addition to daily charts, monthly energy charts are also produced.
- Option to disable the export of daily charts.

Note:

- Custom simulation allows the selection of a weekly building operation pattern (e.g., 5/7 days = the building operates according to the heating settings and user presence for 5 days and remains inactive for 2 days).
- Desired indoor temperatures for heating and cooling.

4. Occupancy and number of occupants



The user schedule is set by left-clicking the boxes, which changes their color. The colors represent user presence and absence.

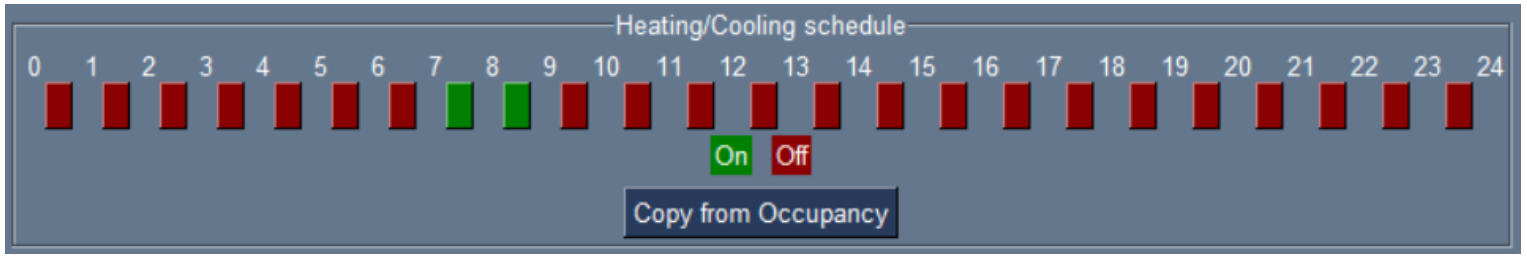
In the example, the schedule corresponds to:

- User presence: 00:00–01:00
- User absence: 01:00–23:59

The number of occupants during the presence schedule is also entered.

Note: At present, occupants are assumed to perform activities typical of residential and office buildings (60 W per person).

5. Heating/Cooling schedule maximum heating/cooling power, mechanical ventilation schedule

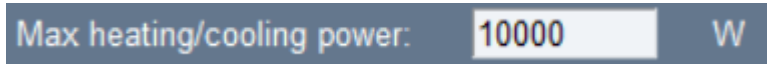


The heating/cooling schedule is set by left-clicking the boxes, which changes their color. The colors represent the availability or unavailability of heating and cooling.

In the example, the schedule corresponds to:

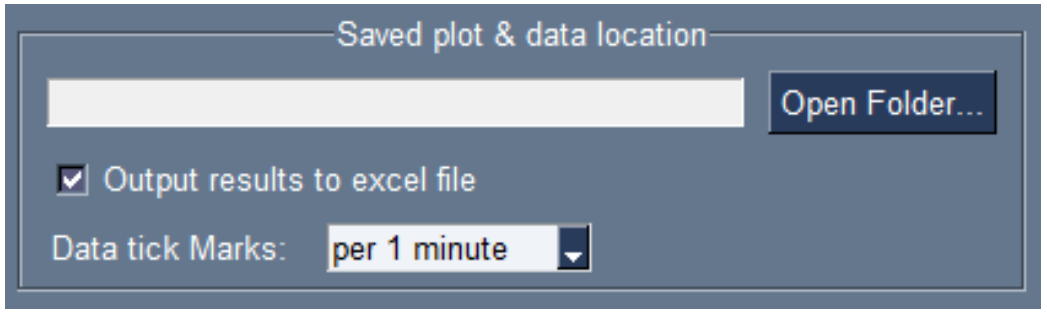
- Heating/Cooling available: 07:00–09:00
- Heating/Cooling not available: 00:00–07:00, 09:00–23:59

Note: Heating/cooling always operates according to the setpoint temperatures during the scheduled operating hours.



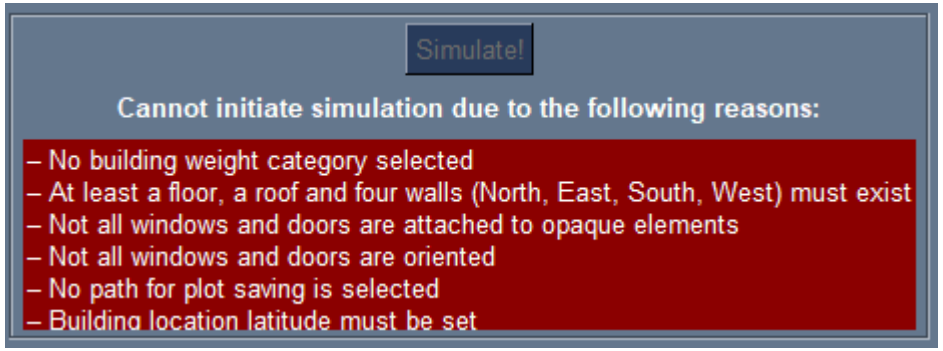
The maximum available heating/cooling power at any given time is also entered, in watts (W).

6. Select result data saving location



The selection of the location follows a procedure similar to that of Step 3, except that this time a location is selected (e.g., a folder, desktop, etc.).

If the option to export an Excel file with the results is selected, the results time step can be set to 1 minute or 1 hour.



To start the simulation, all steps must have been completed correctly so that the check can confirm that the software has all the necessary data.

The “Simulate!” button starts the simulation. The results are generated in the location specified by the user.